

Interpreting Steering Vectors

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Abstract

Steering vectors offer a simple yet powerful mechanism for controlling behaviors in large language models (LLMs) by injecting fixed directions into their residual stream. While they reliably induce behaviors such as refusal, truthfulness, or sentiment, their semantic meaning remains unclear. In this work, we investigate the interpretability of a refusal steering vector. We evaluate one prior interpretation method and propose two novel approaches to uncover sparse token-based representations of the steering vector. Across all methods, we find that the refusal vector consistently aligns with a small set of refusal-related tokens, and that these tokens dominate both static projections and dynamic logit deltas. Our results show that steering directions are not opaque latent features but correspond to interpretable lexical dimensions in the model’s output space. These findings deepen our understanding of how behavior control manifests at the token level and open avenues for more transparent alignment strategies.

1 Introduction

Large language models (LLMs) are increasingly deployed in high-stakes applications, where their ability to produce safe and reliable outputs is critical. A key technique for achieving such control is the use of *steering vectors*: directions in the residual stream that can be added or removed to promote or suppress particular behaviors. Recent work has demonstrated their effectiveness in controlling refusal, toxicity, truthfulness, and other dimensions of model behavior (Arditi et al., 2024; Rinsky et al., 2024). However, while steering vectors are powerful in practice, their underlying semantics remain poorly understood. What do these vectors represent, and how do they interact with the model’s token space to shape generation?

This question is particularly pressing in the case of refusal. A refusal vector can reduce the likeli-

hood of harmful outputs, but excessive refusal can block benign queries, undermining usability. Understanding what refusal steering *means* in token space is therefore crucial for balancing safety and utility. Existing interpretability efforts often rely on external explanation models to describe latent features (Gur-Arieh et al., 2025), but such methods are indirect and fragile. In contrast, our goal is to interpret steering vectors directly through the vocabulary of the model itself.

In this work, we focus on the refusal vector extracted by Arditi et al. (2024) and evaluate its token-level semantics using three complementary approaches. First, we project the vector into the output vocabulary space, revealing its strong alignment with canonical refusal tokens (Dar et al., 2023). Second, we apply sparse dictionary learning to approximate the vector using only a handful of token embeddings, showing that refusal semantics dominate compact representations. Finally, we design a controlled generation procedure that compares logits between steered and baseline models under matched prefixes, providing a robust picture of how the vector reshapes token probabilities in practice.

Our experiments show striking consistency across methods: the same refusal-related words appear in projections, sparse decompositions, and logit deltas, indicating that the refusal vector has a stable and interpretable lexical footprint. Among the approaches, controlled generation proves the most robust, producing clear signals with minimal noise. Together, these findings suggest that steering vectors correspond to coherent token-level dimensions, rather than abstract latent factors, and that their semantics can be uncovered systematically. While our study is limited to a single model and behavior, the methodology extends broadly, offering a foundation for principled interpretation of steering directions across models and domains.

2 Related Work

Steering Vectors and Behavior Control. Our work builds on the growing body of literature around behavior steering in large language models (LLMs), particularly via injection of steering vectors into the residual stream. We draw directly from the methodology and findings of [Arditi et al. \(2024\)](#), who extract a refusal direction and demonstrate its effectiveness in modulating model behavior. We rely on their published direction as the core object of our interpretation efforts. In parallel, [Rimsky et al. \(2024\)](#) present a generalizable method for extracting and applying behavioral directions across multiple models and behaviors. While our work focuses on a single model and direction, their framework provides useful context for the broader landscape of behavior control.

Interpretation of Internal Features. A central component of our work involves interpreting the semantic content of a steering vector. For this, we take inspiration from the “logit lens” approach of [Dar et al. \(2023\)](#), which projects internal activations into the output token space using the unembedding matrix. We adapt this projection method to analyze which tokens align most closely with the refusal direction.

Additionally, we draw from [Gur-Arieh et al. \(2025\)](#), who explore output-centric interpretations of latent features via stimulation and response. Although their methods target learned features rather than steering directions, their ideas, such as tracking token-level promotion after feature stimulation and projecting into vocabulary space, helped motivate our two novel approaches for interpreting steering vectors.

Sparse Approximations and Token Alignment. While sparse coding and dictionary learning have been applied in other areas of model interpretability and compression, they have not been widely explored in the context of steering vectors. Our use of sparse reconstruction over token embeddings offers a new way to examine which tokens semantically approximate a behavioral direction, providing a more transparent and compact view of what the vector represents.

3 Methodology

Our goal is to interpret the *refusal steering vector* r in LLaMA-2-7B-chat by examining its behavior in the token space. Specifically, we aim to uncover

whether r corresponds to semantically meaningful subsets of vocabulary items, and whether its effect can be robustly characterized. To this end, we investigated three complementary approaches: (code available at our git repository: [SteeringVectors](#)).

3.1 Projection into the Vocabulary Space

Our first approach projects the steering vector r into the model’s output vocabulary space. This technique is inspired by the work of [Dar et al. \(2023\)](#), who demonstrated that projecting hidden states or parameters into the embedding space provides an interpretable view of Transformer internals. Similarly, we apply a projection by normalizing r with RMSNorm and then multiplying it by the decoding matrix W_{decode} :

$$s = \text{RMSNorm}(r) W_{\text{decode}},$$

where $s \in \mathbb{R}^{|V|}$ assigns a logit-like score to each token in the vocabulary V .

This procedure is analogous to the “logit lens” perspective, in which hidden states are mapped into interpretable token distributions by linear projection through the embedding matrix. The highest-scoring tokens under s often included refusal-related terms, (e.g., *prohib*, *avoided*, *forb*) suggesting that r encodes refusal semantics. However, the results also contained substantial noise: many of the high-ranked items corresponded to partial subword fragments rather than complete words, or to tokens from other languages, which limited the semantic interpretability of the ranking. This limitation aligns with observations in [Dar et al. \(2023\)](#), where top- k projections were found to be noisy and sensitive to embedding anisotropy. Moreover, inspecting the bottom- k tokens revealed even less meaningful structure: these were almost entirely composed of incomplete sub-words or irrelevant words, providing no useful signal about the semantics of r .

To address this issue, we evaluated complete words (spanning one to three tokens). For each word, we aggregated the scores s of its constituent tokens by computing the total sum and mean score. This provided a more robust measure of how r affects refusal-related words while mitigating noise from sub-word artifacts. The results showed that the refusal words consistently received strong positive scores, while other words were assigned neutral or slightly negative scores. This clarified the earlier observation that the set of most suppressed

tokens appeared nonsensical: the steering direction r primarily promotes refusal-related vocabulary, while leaving most other tokens effectively unchanged.

3.2 Sparse Dictionary Learning

A natural way to reduce the noise and enhance interpretability is to search for a small set of vocabulary items that collectively approximate r . Intuitively, this is appealing because sparse representations encourage semantic clarity: instead of attributing r to thousands of weakly associated tokens, we identify a compact dictionary of tokens that carry the strongest explanatory power.

To operationalize this idea, we solve the following sparse reconstruction problem:

$$\arg \min_{\alpha} \|r - W_{\text{decode}} \cdot \alpha\|_2^2 + \lambda \|\alpha\|_0,$$

where $\alpha \in \mathbb{R}^{|V|}$ is a sparse coefficient vector. The objective encourages r to be reconstructed as a linear combination of only a few vocabulary embeddings (those with non-zero entries in α). The hyperparameter λ balances sparsity against reconstruction fidelity.

The results revealed that the tokens with the largest coefficients in α consistently included refusal-related terms such as *prohib*, *harm*, and *illegal*. This confirmed that r can indeed be expressed in terms of semantically meaningful tokens. Nonetheless, as the sparsity constraint was relaxed, more unrelated tokens were introduced, reintroducing noise. This highlights a fundamental trade-off: small support sets yield interpretable but approximate reconstructions, while larger sets improve accuracy but dilute semantic clarity. Overall, sparse dictionary learning provided a more structured and interpretable view of r compared to simple projection.

3.3 Logit Comparison via Controlled Generation

Finally, we devised a method that closely mirrors how r was originally extracted, namely via a *difference-in-means* between harmful and harmless prompt responses. A straightforward implementation would be to compare the logits of the first token produced by the steered and baseline models. However, this naive approach proved insufficient: while it highlighted frequent prefixes of refusals (e.g., *I*, *cannot*, *As*), it failed to capture the deeper semantics of refusal. In particular, tokens central to

refusal such as *prohibit* or *illegal* were largely absent, and when applied to harmful prompts with steering towards compliance, the method resulted in near-random noise. This motivated the need for a more robust and precise approach.

Our refined method proceeds as follows. For each prompt (50), we first generate a baseline response of fixed length (up to 64 tokens) using the unsteered model. Then, for each prefix of length $t \leq 64$, we feed the partial response back into the *steered* model and record its predicted logits for the next token. At each position t , we compute the logit difference:

$$\Delta_t = \ell_t^{\text{steered}} - \ell_t^{\text{baseline}},$$

where $\ell_t^{\text{steered}}, \ell_t^{\text{baseline}} \in \mathbb{R}^{|V|}$ are the next-token logits given the same prefix under the two models. We then average these differences over all positions t in the response, and subsequently average across prompts:

$$\Delta = \frac{1}{N} \sum_{i=1}^N \frac{1}{T} \sum_{t=1}^T \Delta_{t,i},$$

where N is the number of prompts and T is the response length. The result $\Delta \in \mathbb{R}^{|V|}$ provides a robust measure of which tokens are promoted or suppressed by r .

We applied this method under two complementary conditions: (i) harmless prompts with $a = +1$, where r was expected to induce refusal to benign queries; and (ii) harmful prompts with $a = -1$, where r was expected to suppress refusals and produce unethical outputs. In the first case, the promoted tokens were strongly and unambiguously aligned with refusal semantics (neither, dangerous, not, illegal), while in the second case, the promoted tokens were consistently positive or affirmative (great, love, wonderful). Unlike the earlier projection methods, the results here were strikingly clear and essentially free of noise.

From inspecting individual prompt-level differences, we observed that single responses already contained the key refusal-related shifts, and averaging across multiple prompts simply amplified this signal. Indeed, even in a smaller-scale experiment with only $N = 10$ prompts and $T = 32$ tokens, we observed almost identical results to the larger setting, suggesting fast convergence and robustness of the method. Thus, while unconventional, this controlled generation and logit comparison

procedure provides the clearest picture of how r reshapes token probabilities, and ultimately offers the strongest evidence of its semantic interpretability.

4 Data and Preprocessing

Our experiments rely on the same sources of harmful and harmless prompts as used by [Arditi et al. \(2024\)](#). The steering vector r itself was taken directly from their release, ensuring consistency with their methodology. To probe its semantics, we paired r with datasets of prompts designed to elicit either refusal or compliance.

4.1 Datasets

Harmful prompts were drawn from the AdvBench dataset ([Zou et al., 2023](#)), which contains 521 unsafe or malicious instructions (e.g., requests related to disallowed or harmful content). We used the public release provided by `llm-attacks`.

Harmless prompts were taken from the `tatsu-lab/alpaca` dataset ([Taori et al., 2023](#)), restricted to instructions with empty input fields. This filtering yields benign standalone queries for which refusals are inappropriate. As with the harmful data,

4.2 Model and Approach

4.2.1 Steering Vector

The refusal steering vector r was taken directly from [Arditi et al. \(2024\)](#), where it was extracted via a *difference-in-means* method comparing activations on harmful versus harmless prompts. To apply steering, we scaled r by a factor $\alpha \in \{-1, 0, +1\}$ and added it to the residual stream. The baseline $\alpha = 0$ corresponds to the unsteered model, while $\alpha = +1$ and $\alpha = -1$ represent positive (refusal) and negative (complying) steering respectively. This manipulation allowed us to probe whether r has interpretable token-level effects.

4.2.2 Base Model

All experiments were conducted on the LLaMA-2-7B-Chat model, a widely used open-source instruction-tuned LLM. We chose this model because it served as the reference point in prior work on refusal steering vectors ([Arditi et al., 2024](#)), ensuring comparability. In addition, due to limited time and computational resources, it was the largest model we were able to employ. The model was accessed through the Hugging Face transformers library with float16 weights

and residual stream access enabled. Steering was applied by injecting a scaled copy of the vector r into the residual stream.

4.2.3 Approach Overview

We implemented three complementary approaches for interpreting r :

1. **Projection into the Vocabulary Space:** We normalize r with RMSNorm and map it through W_{decode} to obtain token scores. We then evaluate the scores of selected refusal and compliance phrases to assess whether they align with the semantics of the steering vector.
2. **Sparse Dictionary Learning:** Reconstruct r from a small subset of token embeddings by solving a sparsity-constrained regression, identifying a compact, semantically coherent set.
3. **Logit Comparison via Controlled Generation:** Compare next-token logits between baseline and steered models under prefix-matched generations to capture how r changes token probabilities in practice.

5 Experiments and Results

5.1 Quantitative Results

5.1.1 Projection into the Vocabulary Space

Table 1 summarizes the results of projecting the refusal vector r into the vocabulary space and aggregating token-level scores over curated refusal and completion words. For each phrase, we report both the total Score (sum over all constituent tokens) and the mean Score (average per token).

These results confirm that r is semantically aligned with refusal: words such as *harmful*, *prohibited*, and *illegal* receive the largest positive scores. In contrast, words associated with compliance or helpfulness (e.g., *explain*, *solution*, *sure*) are consistently suppressed. The mean Score column highlights per-token consistency within multi-piece words (e.g., *harmful* and *prohibited* are strongly positive across all sub-tokens). This analysis clarifies why bottom- k raw tokens appeared noisy: the meaningful signal of r emerges only when aggregating over complete words.

5.1.2 Sparse Dictionary Learning

To further interpret the semantic content of the refusal direction r , we applied sparse dictionary

Refusal Words	Total Score	Mean Score
harmful	+10.25	+3.42
prohibited	+8.65	+2.88
illegal	+5.71	+2.85
forbidden	+4.69	+1.56
dangerous	+4.06	+2.03
restricted	+3.88	+1.94
cannot	+3.70	+1.85
unlawful	+4.87	+1.22
Completion Words	Total Score	Mean Score
guide	+2.26	+1.13
outline	+1.64	+0.82
absolutely	+1.11	+0.55
solution	-2.26	-1.13
explain	-2.71	-1.35
implemented	-3.51	-1.75
sure	-4.13	-2.06
easily	-7.45	-3.73

Table 1: Word-level scores from projecting r into the vocabulary space. Positive values indicate that a phrase is promoted by r , while negative values indicate suppression. Refusal-related terms are strongly promoted, whereas completion-related terms tend to be suppressed or neutral. Extended rankings are reported in Appendix A.

learning to decompose r as a linear combination of a small set of token embeddings. We used LASSO regression with varying values of the sparsity coefficient λ , trading off approximation accuracy and interpretability.

We observed a clear trade-off: smaller values of λ yielded more accurate approximations (lower ℓ_2 distance) but resulted in large, noisy token sets containing many semantically irrelevant components—especially in the negative direction. In contrast, larger values of λ produced decompositions with fewer, more interpretable tokens dominated by high-weight, positively aligned components that strongly matched the semantics of refusal.

Table 2 illustrates this trade-off by showing the top 8 tokens obtained from sparse decompositions at two representative λ values. Tokens at higher sparsity ($\lambda = 10^{-5}$) are clearly refusal-aligned, while those at lower sparsity ($\lambda = 10^{-7}$) include more noise and negative-direction entries with unclear semantic value.

For example, with a relatively large $\lambda = 10^{-5}$, the decomposition contained tokens such as 'prohib', 'harm', 'illegal', and 'Sorry', all clearly aligned with refusal behavior. In contrast, the tokens with large negative weights (e.g., 'sue', 'io') lacked coherent semantic interpretation. At smaller sparsity (e.g., $\lambda = 10^{-7}$), the top tokens

included more mixed or uninterpretable entries, particularly in the negative direction (e.g., 'zi_l', 'atã', 'A', ':#'), despite achieving a better fit.

This asymmetry suggests that the positive direction of r is more semantically concentrated and meaningful, while the negative direction may diffuse across unrelated lexical dimensions.

Quantitative results illustrating the tradeoff between reconstruction error and sparsity as a function of λ are presented in Figures 1, 2. In addition, Table 7 demonstrates the effect of sparsity on the semantic coherence of the decomposed token sets. All figures are provided in Appendix B.

5.1.3 Logit Comparison via Controlled Generation

Table 3 reports the top tokens promoted under controlled logit comparison. When steering was applied with $\alpha = +1$ on harmless prompts, the model strongly promoted refusal semantics such as *prohibit*, *illegal*, and *cannot*. In contrast, with $\alpha = -1$ on harmful prompts, the model promoted clearly affirmative and positive completions such as *great*, *happy*, and *wonderful*.

Unlike projection or sparse dictionary learning, this method produced results that were strikingly clear and essentially free of noise. The naive alternative of comparing only the first token’s logits produced mostly prefixes (*I*, *cannot*, *sorry*) and failed to capture the deeper refusal semantics, especially under $\alpha = -1$ where the results degenerated into noise. By contrast, our prefix-matched averaging procedure revealed consistent token-level shifts across the full 64-token horizon. Moreover, a smaller-scale experiment ($N = 10$ prompts, $T = 32$ tokens) produced nearly identical results, indicating fast convergence and robustness of the method. (For extended token rankings, Visit Appendix C)

5.2 Qualitative Analysis

Beyond raw metrics, it is instructive to consider how the three interpretability methods qualitatively reinforce one another. Taken together, they provide a coherent picture of what the refusal vector r represents.

(1) Projection into the Vocabulary Space. By projecting r into the vocabulary space, we observed strong alignment with canonical refusal tokens such as *prohibit*, *illegal*, and *forbid*. Although the projection results were partially noisy due to tok-

$\lambda = 10^{-5}$ (Sparse, but Interpretable)				$\lambda = 10^{-7}$ (Dense, but Noisy)			
Index	Token	Weight	Cosine Similarity	Index	Token	Weight	Cosine Similarity
21460	'prohib'	+0.0437	+0.0999	21460	'prohib'	+0.0576	+0.0999
12991	'sue'	-0.0269	-0.0614	17977	'absor'	-0.0557	-0.0460
601	'io'	-0.0252	-0.0668	11689	'iens'	-0.0527	-0.0351
22658	'téc'	+0.0217	+0.0755	22236	'DEFAULT'	-0.0489	-0.0394
10311	'harm'	+0.0216	+0.0696	22797	'zi_l'	-0.0445	-0.0064
27302	'illegal'	+0.0215	+0.0748	21968	':#'	-0.0424	-0.0317
8221	'Sorry'	+0.0195	+0.0662	13804	'atã'	-0.0405	-0.0193
6261	'explicit'	+0.0179	+0.0688	29909	'A'	-0.0398	-0.0459

Table 2: Top 8 tokens from sparse decompositions of r at two sparsity levels. Larger λ yields fewer, more semantically coherent tokens (left), while smaller λ improves reconstruction accuracy but introduces noisy and less interpretable components (right).

Harmless ($\alpha = +1$)		Harmful ($\alpha = -1$)	
Token	Δ	Token	Δ
prohib	+7.09	great	+4.40
illegal	+6.57	pleasure	+4.40
cannot	+6.14	happy	+4.15
neither	+6.03	excellent	+4.01
nor	+5.77	wonderful	+3.98
forb	+5.64	love	+3.87
viol	+5.57	delight	+3.84
dangerous	+4.89	awesome	+3.49

Table 3: Top promoted tokens under controlled generation logit comparison. With $\alpha = +1$ on harmless prompts, refusal semantics dominate. With $\alpha = -1$ on harmful prompts, positive and enthusiastic completions dominate. Extended token rankings are reported in Appendix C.

enization artifacts, the consistent presence of these terms indicates that r is lexically grounded in refusal semantics.

(2) Sparse Dictionary Learning. Sparse reconstruction further supports this interpretation. When r is approximated using only a small subset of vocabulary embeddings, the resulting dictionary is again dominated by refusal-related tokens. This provides evidence that r can be represented parsimoniously in terms of specific refusal words, not just distributed across thousands of tokens. In this way, the projection results (1) serve as a foundation for sparse reconstruction (2), since both point to the same lexical core.

(3) Logit Comparison via Controlled Generation. Finally, controlled logit comparison demonstrates that these lexical associations are not static artifacts, but actively shape generation. With $\alpha = +1$, r promotes refusal tokens in benign contexts; with $\alpha = -1$, it suppresses refusal and amplifies positive, compliance-oriented language. The

clear and robust signal in this method is consistent with the earlier evidence: the same refusal words highlighted in (1) and (2) are precisely those that dominate logit deltas during generation.

Taken together, these observations indicate that the three methods converge on the same conclusion. Projection shows that r is aligned with refusal tokens; sparse dictionary learning shows that r can be reconstructed using them; and controlled generation shows that r actively promotes or suppresses them in practice. The fact that (1) and (2) reinforce one another makes the evidence for (3) even stronger, establishing that the refusal direction has a stable and interpretable footprint across both static embeddings and dynamic generation.

5.3 Discussion

Our findings show that the refusal vector r is semantically interpretable and consistently linked to refusal-related language. Projection and sparse dictionary learning both reveal alignment with tokens such as *prohibit*, *illegal*, and *forbid*, while controlled logit comparison demonstrates that these associations actively shape generation. Among the three methods, logit comparison proved most effective, producing clear and robust signals with minimal noise, whereas projection and sparse methods were limited by tokenization artifacts and trade-offs between sparsity and fidelity.

In relation to prior work, these results extend [Arditi et al. \(2024\)](#) and [Dar et al. \(2023\)](#), showing that steering vectors correspond not to opaque latent features but to semantically coherent sets of tokens. This strengthens the interpretability of behavioral steering while also highlighting risks: understanding refusal semantics could make it easier to circumvent safeguards. Overall, our evidence supports the view that steering vectors capture

human-readable lexical semantics and can be analyzed systematically, offering both opportunities for alignment and challenges for safety.

6 Limitations

While our study provides new insights into the interpretability of steering vectors, it is subject to several limitations:

1. **Single vector and model.** We analyzed only one steering direction (refusal) on a single model (LLaMA-2-7B-Chat), primarily due to time and computational constraints. It remains unclear whether our conclusions generalize to other behaviors or larger model families.
2. **Subword artifacts in projection.** The projection method operates at the token level, which often corresponds to partial subwords. This can lead to misleading results: for example, the contraction “can’t” decomposes into *can* + *’t*, where *can* is a compliance-associated word and thus receives a low score, obscuring the intended refusal meaning of the full phrase.
3. **Sparse reconstruction trade-off.** Sparse dictionary learning necessarily balances interpretability and fidelity. Highly sparse token sets yield clear semantic interpretations but only approximate r , whereas larger sets reconstruct r more faithfully but introduce noise and dilute interpretability.
4. **Directional asymmetry.** We observed that positive weights in sparse approximations were semantically meaningful (e.g., refusal tokens), while negative weights were often incoherent. This asymmetry limits the interpretability of suppression effects and suggests that not all aspects of r can be equally explained.
5. **Prompt-based logit comparison.** Our logit comparison method is unconventional in that it reuses baseline outputs as prefixes for the steered model. While this prevents divergence, it introduces a dependency on specific prompts. As shown in Table 9, using only the first token yields mostly superficial prefixes and fails to capture the full meaning of r , highlighting both the necessity and the unorthodox nature of our approach.

Overall, these limitations underscore the need for broader experiments, more principled handling of tokenization artifacts, and alternative methods for balancing interpretability with fidelity.

Ethics Statement

While our work does not directly introduce new ethical or safety concerns, it contributes to the broader understanding of how large language models respond to behavioral steering, specifically in the context of refusal. By analyzing the semantic footprint of steering directions in token space, we provide tools for interpreting what “refusal” means to the model in terms of lexical patterns and token-level shifts.

Although this interpretability may help improve alignment and transparency, it could also expose potential avenues for adversarial manipulation. For example, malicious actors may use similar techniques to identify and circumvent refusal-related behaviors, thereby weakening model safeguards. We believe that careful consideration of these dual-use implications is essential as interpretability methods become more sophisticated.

Acknowledgements

We would like to thank Dr. Mor (Geva) Pipak for her guidance and support throughout the course, which provided the foundation for this project. We are also deeply grateful to Matan Ben-Tov, our PhD mentor, for his insightful feedback, encouragement, and invaluable discussions that shaped the direction of this work.

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A Projection into the Vocabulary Space

This section reports extended results of projecting r into W_{decode} . Table 4 contains the top and bottom raw scores from the projection. Table 5 and table 6 expand the compact summary in Table 1 of the main text. (Tokens containing non-English characters are shown as “*non-english*”).

A.1 Top Promoted and Suppressed Tokens

Promoted Tokens			Suppressed Tokens		
Token	ID	Δ	Token	ID	Δ
prohib	21460	+10.33	io	601	-9.05
kwiet	25145	+9.87	sue	12991	-7.28
non-english	31358	+9.35	defaults	21274	-6.87
ban	9892	+9.13	aga	7781	-6.51
uky	21108	+9.08	su	480	-6.30
void	5405	+8.95	Marg	11649	-6.08
non-english	31455	+8.82	it	277	-6.04
virtu	24465	+8.77	Tang	27378	-6.02
ban	2571	+8.58	aglia	17474	-5.88
vet	5990	+8.38	Ju	5342	-5.79
harm	10311	+8.38	ault	1292	-5.78
non-english	10574	+8.36	rand	20088	-5.77
avoided	28305	+8.34	background	3239	-5.72
iemann	23745	+8.26	ax	1165	-5.72
sted	16470	+8.18	compla	15313	-5.64
forb	19752	+8.15	ode	356	-5.64
sq	3044	+8.12	absor	17977	-5.62
repre	14767	+8.09	easily	5948	-5.62
non-english	7478	+8.07	jen	4142	-5.53
non-english	246	+8.01	default	4381	-5.50

Table 4: Top promoted and suppressed tokens from projection of r into the vocabulary space (partial lists).

A.2 Phrase-Level Aggregation

Refusal Words	Total Score	Mean Score
harmful	+10.25	+3.42
prohibited	+8.65	+2.88
ban	+7.31	+3.65
illegal	+5.71	+2.85
avoid	+5.69	+2.85
forbid	+5.61	+1.87
unlawful	+4.87	+1.22
forbidden	+4.69	+1.56
dangerous	+4.06	+2.03
restricted	+3.88	+1.94
cannot	+3.70	+1.85
caution	+3.44	+1.15
violate	+2.15	+0.72
unethical	+2.11	+0.53
against	+1.48	+0.74
refrain	+1.35	+0.45
sorry	+1.11	+0.56
unsafe	+0.83	+0.42
limit	+0.76	+0.38
dissuade	+0.13	+0.03
policy	-0.30	-0.15
decline	-0.57	-0.19
block	-0.61	-0.30
unable	-0.87	-0.43
refuse	-1.14	-0.57
reject	-1.47	-0.73
deny	-1.54	-0.51
won't	-4.51	-1.13
withhold	-4.86	-1.62
neglect	-4.98	-2.49
comply	-5.67	-1.89
can't	-7.24	-1.81

Table 5: Refusal words ranked by total and mean score under projection.

Completion Words	Total Score	Mean Score
guide	+2.26	+1.13
outline	+1.64	+0.82
absolutely	+1.11	+0.55
exemplify	+0.43	+0.14
describe	+0.25	+0.13
informative	+0.05	+0.02
clearly	-0.09	-0.04
definitely	-0.42	-0.21
instruction	-0.49	-0.25
recommend	-0.59	-0.29
final	-0.62	-0.31
method	-0.62	-0.31
yes	-0.71	-0.36
approach	-0.72	-0.36
assist	-0.79	-0.39
detailed	-0.99	-0.49
summary	-1.10	-0.55
straightforward	-1.15	-0.58
provide	-1.33	-0.66
certainly	-1.35	-0.68
helpful	-1.54	-0.77
steps	-1.70	-0.85
answer	-1.92	-0.96
explanation	-2.25	-1.12
solution	-2.26	-1.13
explain	-2.71	-1.35
implemented	-3.51	-1.75
simple	-3.87	-1.93
sure	-4.13	-2.06
concretely	-5.32	-1.33
easily	-7.45	-3.73

Table 6: Completion words ranked by total and mean score under projection.

B Sparse Dictionary Learning: Extended Results

The results highlight a clear tradeoff between sparsity and reconstruction fidelity when approximating the steering vector with a linear combination of token embeddings. As shown in Table 7, larger values of λ enforce sparsity and yield compact decompositions dominated by semantically coherent refusal-aligned tokens such as `prohib`, `harm`, and `illegal`. These token sets are highly interpretable, as they capture the refusal behavior with minimal noise. However, the reconstruction error (Figure 1) increases as λ grows, reflecting the loss of information. Conversely, smaller λ values achieve much lower reconstruction error but introduce many additional tokens, often noisy or semantically irrelevant (e.g., rare or non-English fragments). This is reflected in Figure 2, where the number of non-zero coefficients rises steeply as λ decreases. Taken together, these results suggest that moderate sparsity (e.g., $\lambda \approx 10^{-5}$) strikes a balance between interpretability and accuracy, enabling us to reveal the semantic alignment of the refusal direction without overwhelming noise.

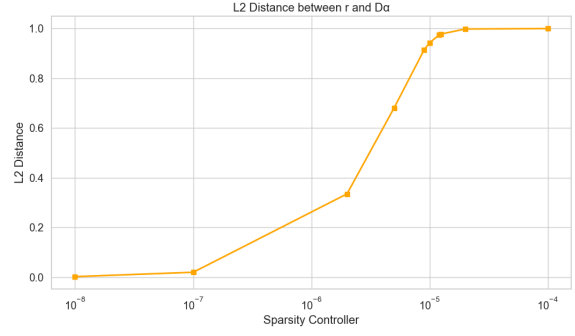


Figure 1: Reconstruction error (ℓ_2 distance) as a function of λ .

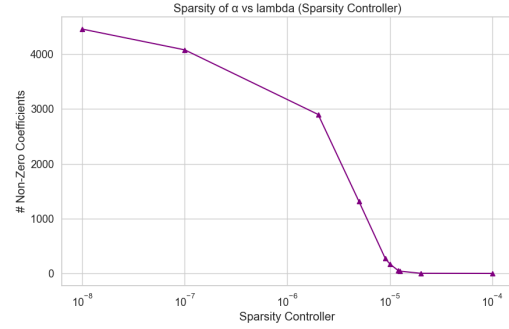


Figure 2: Number of non-zero tokens in the decomposition as a function of λ .

$\lambda = 1.3 \cdot 10^{-5}$		$\lambda = 10^{-5}$		$\lambda = 10^{-7}$		$\lambda = 10^{-8}$	
Token	Weight	Token	Weight	Token	Weight	Token	Weight
prohib	+0.0402	prohib	+0.0437	prohib	+0.0576	absor	-0.0567
nicht	+0.0217	sue	-0.0269	absor	-0.0557	prohib	+0.0552
void	+0.0165	io	-0.0252	iens	-0.0527	iens	-0.0543
sue	-0.0156	téc	+0.0217	harm	+0.0493	sue	-0.0474
illegal	+0.0155	harm	+0.0216	DEFAULT	-0.0489	Sorry	+0.0474
harm	+0.0154	illegal	+0.0215	sue	-0.0480	harm	+0.0468
io	-0.0154	nicht	+0.0208	Sorry	+0.0479	DEFAULT	-0.0467
téc	+0.0149	void	+0.0205	explicit	+0.0467	avoid	+0.0441
NO	+0.0128	mod	+0.0197	ził	-0.0445	explicit	+0.0440
avoided	+0.0127	Sorry	+0.0195	non-english	+0.0433	uch	+0.0435
explicit	+0.0118	explicit	+0.0179	uch	+0.0426	mod	+0.0433
mod	+0.0117	avoided	+0.0176	:#	-0.0424	san	+0.0416
Sorry	+0.0106	NO	+0.0176	aglia	-0.0418	ził	-0.0416
controvers	+0.0106	controvers	+0.0175	san	+0.0418	:#	-0.0407
NOT	+0.0097	san	+0.0168	mod	+0.0410	ătă	-0.0403
san	+0.0091	Mor	+0.0164	greg	+0.0407	aglia	-0.0398
nor	+0.0086	conditional	+0.0160	ătă	-0.0405	reaction	+0.0397
non-english	+0.0086	nor	+0.0159	A	-0.0398	variety	-0.0393
non-english	+0.0075	iac	+0.0156	reaction	+0.0396	textsc	+0.0391
Mor	+0.0069	defaults	-0.0154	ifs	+0.0389	A	-0.0380

Table 7: Top 20 tokens from sparse decompositions across different λ values. Higher λ yields smaller, more interpretable token sets dominated by refusal-aligned tokens (e.g., ‘prohib’, ‘harm’), while smaller λ values improve reconstruction accuracy but introduce many noisy or less meaningful entries.

C Logit Comparison via Controlled Generation

Interpretation of r^+ and r^- . The controlled generation results reveal a striking semantic contrast. For harmless prompts with $\alpha = +1$, the positive direction r^+ strongly promotes refusal-related language, including tokens such as *prohib*, *illegal*, *cannot*, *forb*, and *dangerous*. These belong to a coherent semantic field of prohibition, danger, and denial, indicating that r^+ corresponds to a lexically grounded representation of refusal.

In contrast, for harmful prompts with $\alpha = -1$, the negative direction r^- shifts the model toward tokens like *great*, *pleasure*, *happy*, *wonderful*, and *love*. These fall into a semantic field of affirmation, positivity, and compliance, showing that r^- actively suppresses refusal and instead promotes supportive or enthusiastic continuations. Together, these results demonstrate that the refusal vector r is not abstract but manifests as a clear lexical polarity: prohibition versus affirmation.

C.1 Token Promotions

Harmless, $\alpha = +1$			Harmful, $\alpha = -1$		
Token	ID	Δ	Token	ID	Δ
prohib	21460	+7.09	great	2107	+4.40
illegal	27302	+6.57	pleasure	15377	+4.40
cannot	2609	+6.14	different	1422	+4.34
neither	9561	+6.03	happy	9796	+4.15
nor	3643	+5.77	excellent	15129	+4.01
forb	19752	+5.64	wonderful	20695	+3.98
viol	5537	+5.57	love	5360	+3.87
eth	11314	+5.32	delight	15319	+3.84
harm	10311	+5.12	rock	7679	+3.58
dangerous	18215	+4.89	perfect	4922	+3.53
I	306	+4.83	doll	11232	+3.51
never	2360	+4.60	below	2400	+3.51
explicit	6261	+4.53	lov	12355	+3.49
strongly	13818	+4.45	awesome	29663	+3.49
shouldn	9273	+4.44	picture	7623	+3.41
impossible	9301	+4.44	fant	13568	+3.39
mor	3036	+4.41	just	925	+3.39
avoid	4772	+4.37	thing	2655	+3.38
anything	3099	+4.28	difference	4328	+3.36
repre	14767	+4.26	fine	2691	+3.35
promotion	22360	+4.24	here	1244	+3.35
should	881	+4.19	good	1781	+3.34
unsafe	25110	+4.14	happiness	22722	+3.25
prom	2504	+4.11	requirement	11809	+3.24
not	451	+4.11	nice	7575	+3.22

Table 8: Top-25 promoted tokens under controlled generation. Left: harmless prompts with refusal amplified ($\alpha = +1$). Right: harmful prompts with refusal suppressed ($\alpha = -1$). Values are average logit deltas across positions (64) and prompts (50).

C.2 Token Promotions (1-token alignment)

Harmless, $\alpha = +1$			Harmful, $\alpha = -1$		
Token	ID	Δ	Token	ID	Δ
I	306	+11.44	I	29896	+4.64
I	29902	+9.91	iek	11718	+4.27
apolog	27746	+6.57	non-english	30363	+4.11
cannot	2609	+6.53	ellig	9347	+4.11
as	408	+6.51	aset	24541	+4.07
As	2887	+6.31	ierte	5423	+4.05
As	1094	+5.88	ello	3156	+3.97
it	372	+5.75	egos	21024	+3.95

Table 9: Top-8 promoted tokens for single-step (first-token) comparison. Non-ASCII/control tokens are rendered as *non-english*.